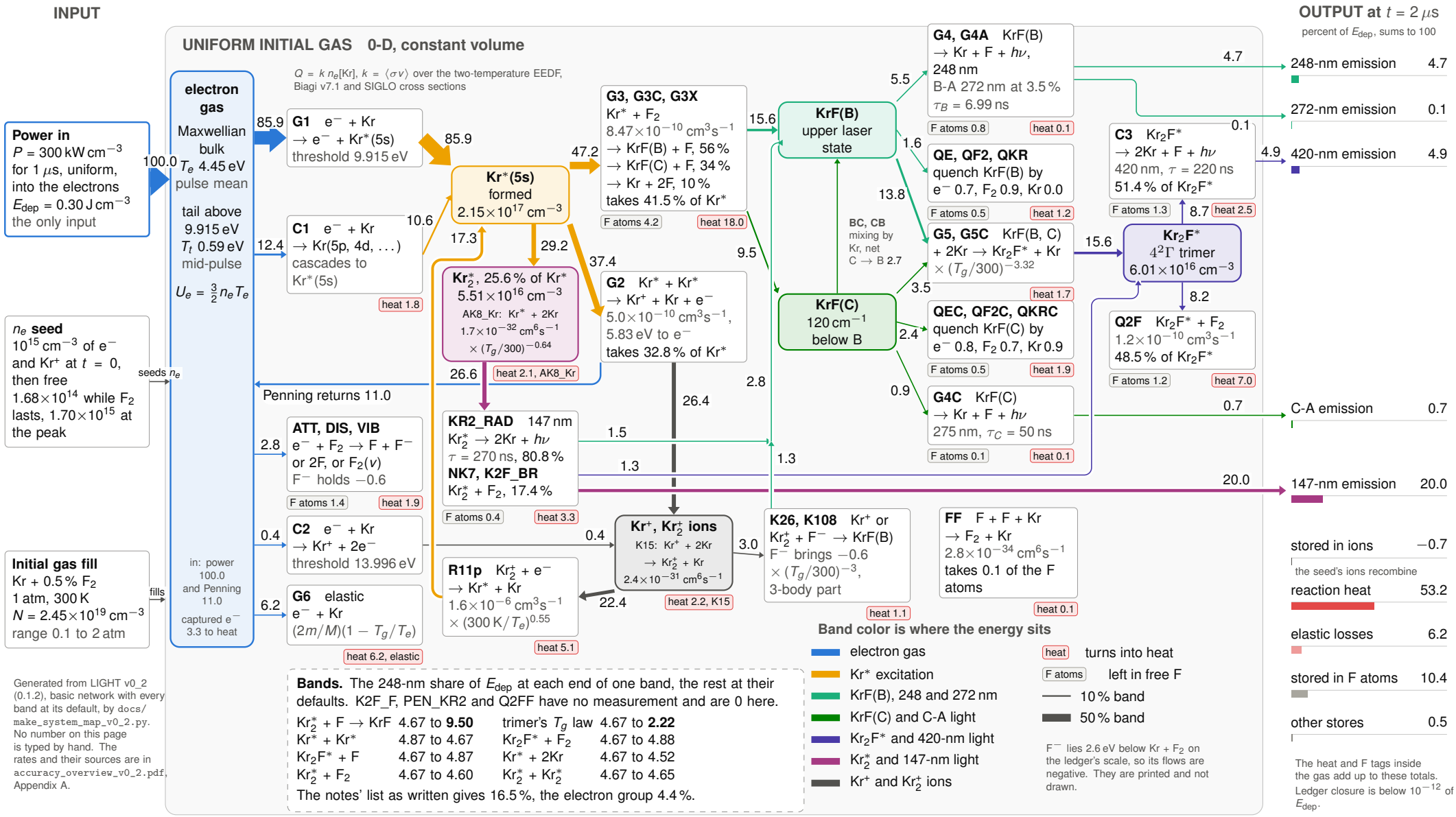
# LIGHT system map, v0_2: energy flow through the Kr/$F_2$ kinetics

Reference point: 1 atm of Kr with 0.5 % $F_2$ at 300 K, $P$ = 300 kW cm$^{-3}$ for 1 $\mu$s so that $E_{dep}$ = 0.30 J cm$^{-3}$, $n_e$ seeded at $10^{15}$ cm$^{-3}$ and then free, every band at its default, run to 2 $\mu$s. Band width is proportional to the energy flow. Numbers are percent of $E_{dep}$, rounded to 0.1.




The notes' outputs as flows, which overlap: to Kr* 98.3 %, to Kr+ 26.8 %, to KrF(B) and KrF(C) 23.8 %, heat 53.2 %, elastic losses 6.2 %, emission 30.4 %. Of the Kr* formed, 25.6 % forms $Kr_2^*$ (15.1 % of the Kr* used before the $F_2$ burns out), and 80.8 % of that $Kr_2^*$ radiates at 147 nm. Of the ions that recombine, 88.7 % meet an electron and return to Kr*, and 11.3 % meet $F^-$ and form KrF(B). Of the KrF formed, $8.93\times10^{16}$ cm$^{-3}$, 4.7 % came from ion-ion recombination and 5.4 % from $Kr_2^* + F_2$. Of it 19.6 % radiates at 248 nm, 0.7 % at 272 nm and 3.3 % in the C-A band, 62.0 % forms the trimer, and electrons, $F_2$ and Kr quench 5.4 %, 5.7 % and 3.4 %. At the end of the run $T_g$ = 649 K and $p$ = 2.18 atm, and $F_2$ fell below 10 % of the fill at 640 ns.